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Task Description:	Adopting hybrid reality (HR) technology still has many of the same challenges as virtual reality (VR). Namely, there is little to no data about the validity of evaluations conducted with HR mockups, an established process, or criteria for evaluating the appropriateness of HR mockups. Because verifications are final and only need to happen once, HR mockups must be adequately vetted before being approved for use in verifications. Further hindering the adoption of these technologies is the lack of consensus on what constitutes a high-fidelity extended reality (XR) mockup. Even for physical mockups, there are guidelines and common criteria, but nothing has been crystallized (Silva-Martinez et al., 2022). Discussion about when an XR mockup can be used would be greatly helped by clearly defining what a high-fidelity XR mockup is and how it might be measured. The proposed work builds upon Phase 1 by focusing on HR technology and continuing the work (e.g., XR for Verification Framework) started in Phase 1. [Ed. Note: for the Phase 1 investigation, see: "AR/VR and Hybrid Reality for Verification Assessments: Phase I" under PI Robertson.]
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This proposal qualifies for directed study because of the time-sensitive nature and scope of the work. Currently, providers are actively petitioning NASA for permission to use hybrid reality in verifications, adding immediacy to the topic. Unfettered access to NASA experts and commercial partners is critical to studying this topic, a level of access external researchers would not have. Finally, conducting this work at NASA provides the researchers access to NASA-representative physical and virtual mockups, maximizing the validity of the research.
Research Impact/Earth Benefits:	The data from this project may will inform future decisions on whether to employ extended reality technologies for verification assessments at NASA. Additionally, this project seeks to generate guidelines for how to assess fidelity and validate extended reality mockups for use in verification assessments. These guidelines may be useful for other domains (e.g., safety and training) and inform future standards applicable to XR mockups for verification assessments.
Task Progress:	NASA providers are using Virtual Reality (VR) and Mixed Reality (MR) technologies more often in their system development process. The use of these technologies, collectively referred to as Extended Reality (XR) technologies, has been limited to the development phase, but there have been requests to integrate VR and MR mockups into verifications. Verifications require, at minimum, a high-fidelity mockup for activities involving human participation (tests, demonstrations, inspections). So, NASA's position has been to limit the use of XR technologies to developmental testing only. However, because XR technologies continue to evolve, allowing for higher fidelity builds, and have the potential to provide cost savings, it is anticipated that NASA providers will continue to explore how these technologies can be used in verifications. There have been many conversations surrounding integrating extended reality (XR) technologies into the verification process. Prior interest was focused on VR tech. However, now that it has matured, requests for consideration have shifted to MR. MR technology shares many of the strengths but may not share some of the critical shortcomings of VR. Specifically, MR supports integration of physical objects, such as a suit, which may be critical to evaluating a system. To better inform conversations regarding the use of XR technologies for verification, this project sought to establish a set of XR mockup development and validation best practices, collect data comparing an MR to a physical reality mockup and mature a framework for evaluating the appropriateness of XR mockups for use in verification. To inform best practices for the evaluation of XR mockups, we performed a literature review. This literature review covered the domains of clinical psychology, civil engineering, gaming, and education/training. The goal was to understand the most important factors for effective use of XR technologies across a breadth of applications by identifying common techniques and measures for evaluating the goodness of XR simulations. To conduct the review, we looked at articles within each domain and identified common measures of quality and methods for evaluating the goodness of XR simulations. From these articles, we surmised the following points. First, there are no standard measures of quality. Fidelity is clearly the most important factor to measure how "good" an XR simulation is. Presence was also a common outcome measure. However, presence is likely a more important factor for applications like games and therapeutic interventions than for usability testing. Nausea was reported in some articles but not all. It may be that nausea is not worth the time it takes to field a formal measure, but instead rely on verbal report from participants. It may also be that nausea measures are underreported because nausea symptoms are rare or not impactful to most participants' experience. Another observation from the literature review was that VR is still the most dominant XR technology, but MR is up and coming. The gap between VR and MR may be due to the maturity of VR technology and the development tools that allow for even novice users to design VR simulations with ease. The barrier to entry for MR may still be too high, and MR will see more adoption when easy-to-use development tools become available. To help guide future adoption of XR mockups for verification, we synthesized literature, existing frameworks, interviews, and verification practices to develop a framework for evaluating the fidelity of XR mockups and validating them for use in verifications. The first step of the framework was to identify and define mockup categories. We elected for a simpler approach to categorizing XR mockups with three categories of Physical Reality (PR; a fully physical mockup), Mixed Reality (MR; a blend of physical and virtual elements to form a mockup), and Virtual Reality (VR; a completely virtual mockup). We believe this should help simplify understanding of what kind of mockup is being discussed and help to guide conversations on what aspects are important to consider relative to the verification goals (e.g., a VR mockup will likely lack haptic fidelity). Within the framework, we define mockup fidelity as "the degree to which a simulation or mockup replicates a real-world system from the user's perspective, with respect to form, function, and user-interaction." Fidelity can be further decomposed into subjective fidelity and objective fidelity. Subjective fidelity is the user's judgment of how similar the mockup/simulation is to its real-world counterparts. An example measure of subjective fidelity would be a questionnaire (e.g., How realistic is the way you pushed the button?). Objective fidelity is the quantifiable degree with which the mockup/simulation replicates the system(s) to be evaluated. For example, how many steps of the verification task are captured by the VR simulation? In addition to the subdimensions explicitly identified in our current definition of fidelity, we identified avatar fidelity and simulation equipment fidelity as two more important aspects of fidelity to be considered. In addition to fidelity, validity was identified as the other important challenge to the adoption of XR mockups for verification. We found several approaches for validating XR mockups, but the applicability and strength of evidence each method produces vary. We recommend verification that equivalence testing provides the highest form of evidence, followed by agreement coefficients and then correlation. Equivalence tests allow for the scoping of equality by setting equivalence boundaries. ICC and correlations are not as specific as equivalence testing in terms of supporting

	equivalencing, but can establish the amount of shared variance in verification outcomes, a common practice is validation literature. Transfer of training and expert-novice comparisons are common validation techniques in training/learning domains, but don't apply well to the verification use-case. The main reason is that this form of validity is nebulous and lacks specificity for stating how "equal" two mockups might be. T-tests are likewise inappropriate for validating an XR mockup, as the t-test is designed to find differences; a lack of a significant finding is not evidence of equality. Within the literature, there is a lack of evidence regarding the equivalence of MR and PR mockups in terms of their appropriateness for conducting verification testing and similar activities (e.g., usability testing. To fill this gap, we conducted an experiment comparing an MR and a PR mockup. In this study, 13 participants completed 4 verification-like tasks in both a physical high-fidelity, physical reality (PR) space vehicle mockup and its mixed reality (MR) equivalent. Data consists of outcomes from the various tasks, such as success (yes/no), time to complete, acceptability, and subjective judgments (e.g., is there enough room to perform exercises?). Overall, the mixed reality (MR) mockup proved to be a highly effective and statistically equivalent substitute for the physical mockup in many critical areas, particularly for overall system usability (NMSUS) and general workload (Bedford). However, the two systems were not equivalent for tasks that appear to rely more heavily on physical interaction or spatial judgment. A secondary finding was that the order in which participants used the systems did not significantly impact the results. Whether a user started with the MR headset or the physical mockup did not systematically alter their performance or ratings. In short, we found that the MR system successfully replicates the physical experience for many procedural and usability-focused tasks, but that physical mockups may remain the benchmark for evaluating tasks that require precise physical feedback or detailed spatial assessment. These results also demonstrated that equivalence testing can be conducted with small samples. Generally, equivalence testing is performed with larger samples (20+), and we thought it may be difficult to perform at NASA, generally, samples for verifications are 10 or fewer. This study demonstrated that equivalence testing is possible with smaller samples, though it may be difficult to get reliable results depending on the outcome measure.
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